Continuous Sensor Health Monitoring and Risk/Capability-aware Graded MRM Supervision for Fail-Safe Operation of Autonomous Vehicles

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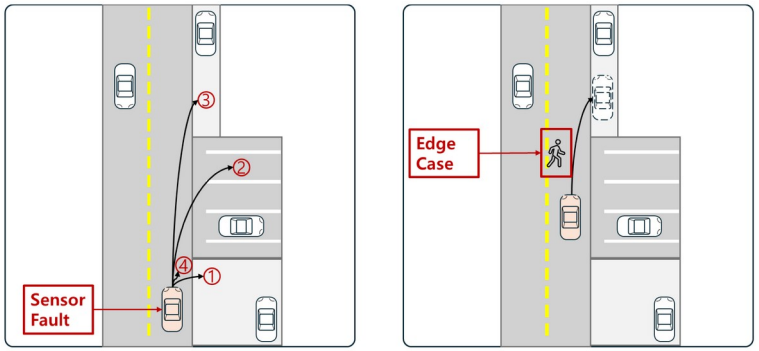
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Outline

- **Background & Problem**
- **Related Work**
- **Proposed System: Health Monitoring & MRM Supervision**
- **Experimental Setup: 1,862-Case Closed Loop**
- **Results: Detection, Encounters, Causal Cap Effect**
- **Conclusion**

Background & Problem

- **Level-4 autonomy must reach a minimal risk condition on its own — ISO 26262 FTTI; SOTIF (ISO 21448)**
- **Evaluated stock stack: in-lane stop only; faults unmonitored → 238–275 m blind driving**
- **Design axes: stop now? / how far? / how fast?**



(a) Refuge options upon sensor fault (b) Edge case — pedestrian dart-out during evacuation

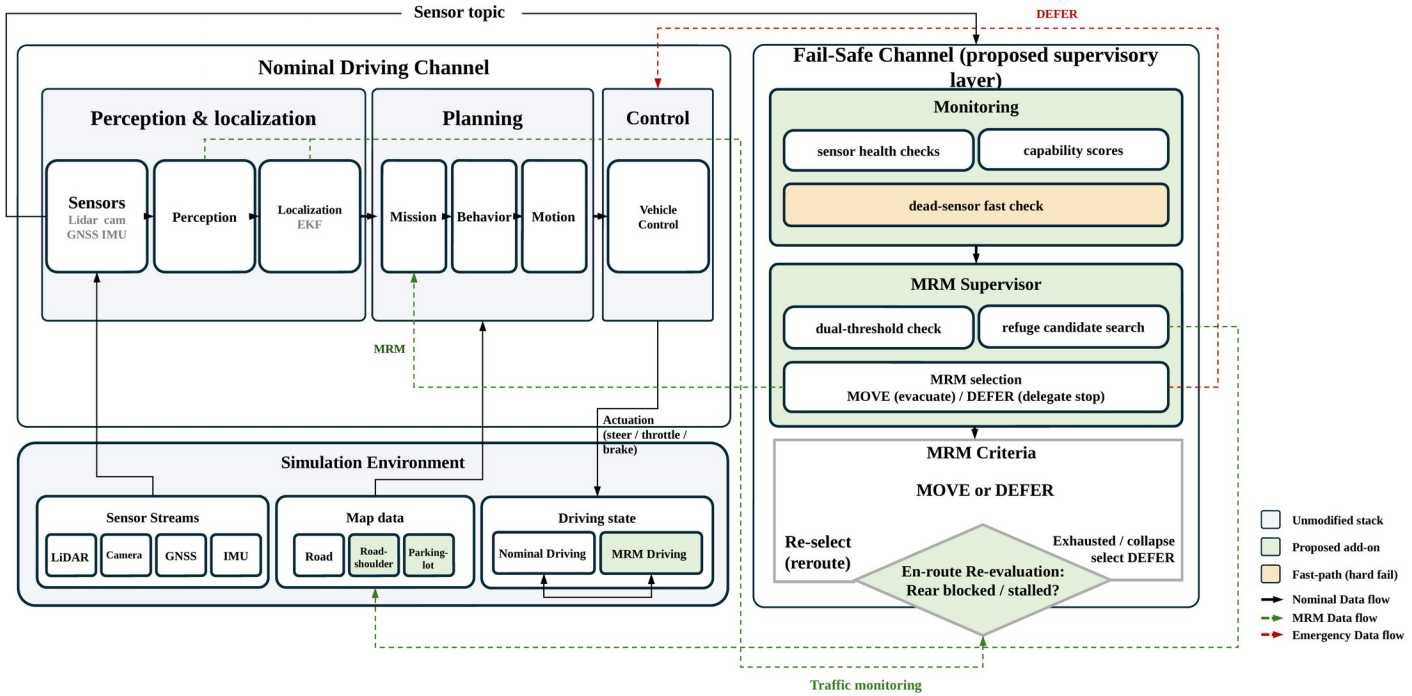
① free space ② parking lot ③ downstream shoulder ④ in-lane stop — candidates ranked by the cost function

Related Work

- **Adaptive-MRM studies presuppose the fault state (Xue 2018, Tang 2026)**
  - **Selection by traffic TTC/THW**
  - **Validation on representative scenarios**
- **This work: estimate faults online as a continuous health score**
  - **Graded MRM selection + evacuation speed cap**
  - **1,862-case closed loop with verified collision verdicts**

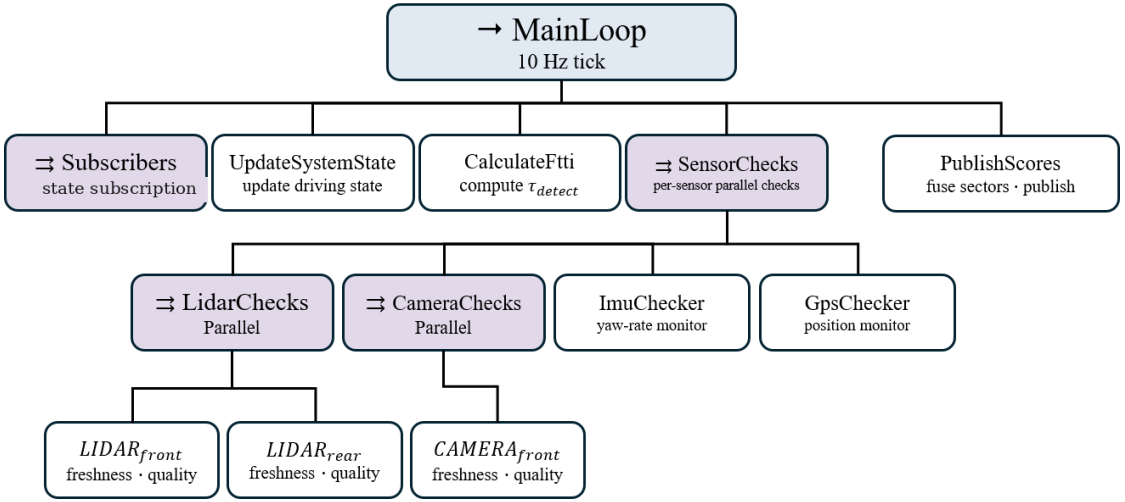
Proposed System

- **Non-intrusive add-on: read-only monitoring, standard interfaces**
- **MOVE (reroute + speed cap) vs DEFER (delegate to stock stop)**



Continuous Health Scoring

- **Freshness · quality · physics scores per sensor at 10 Hz (behavior tree)**
- **8-sector fusion → health = min(perception, localization)**
- **Dual thresholds 0.7 / 0.3 → NORMAL / DEGRADED / CRITICAL**



MRM Supervision & Speed Cap

- **Refuge by cost: exposure time + traffic-light observability gate**
- **Speed cap 3.0 m/s from $d = v\tau + v^2/2a$ (model bound ≈ 3.9 m/s)**
- **Hard fails: evidence-aware fast path decides within 1.5 s**

Monitored Condition	Continue driving	Shoulder stop	Parking-lot stop	In-lane stop	Emergency stop
NORMAL $(health > \theta_H)$	●	x	x	x	x
DEGRADED $(\theta_L < health \leq \theta_H)$ *or dead sensor*	x	●	●	○	x
CRITICAL $(health \leq \theta_L)$ *or no candidate*	x	x	x	●	○

● selected (cost-ordered) ○ fallback on move failure x not selected

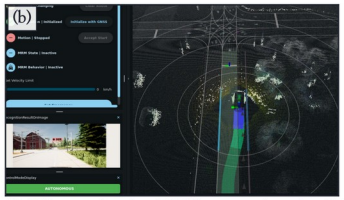
θ_H : intervention-entry (upper) threshold θ_L : minimum-capability (lower) threshold

Experimental Setup

- **CARLA 0.9.16 ↔ Autoware 0.39.0 closed loop (AWCL), Town07, 15 km/h**
- **1,862 valid cases: fault × severity × traffic × method + encounters**
- **Full simulator restart every case — independence guaranteed**



(a) Town07 driving scene (ego vehicle)



(b) RViz perception view (LiDAR, camera, planned path)



(c) custom vector map with shoulder/parking-lot candidates



(d) shoulder-evacuation MRC reached (traffic passes)



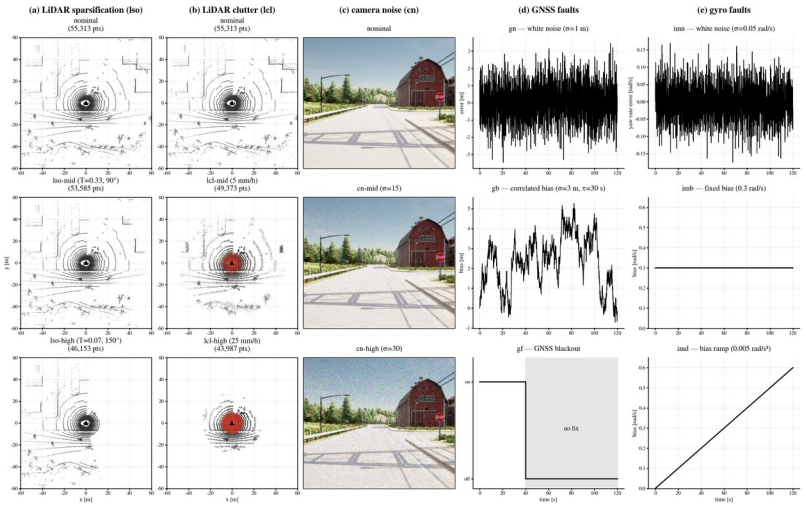
(e) parking-lot evacuation MRC (aerial view)



(f) pedestrian dart-out (crossing from right lane edge)

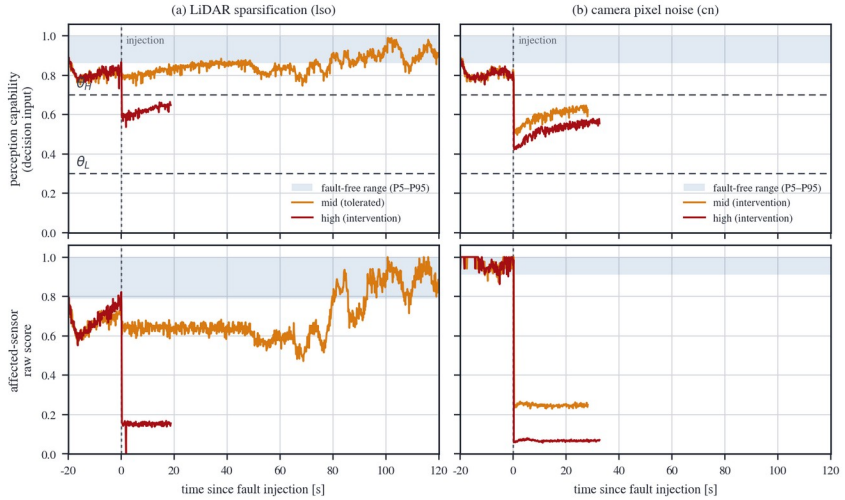
Fault Model & Verdict Reliability

- **11 physics-based fault types (LiDAR, camera, GNSS, IMU)**
- **Pedestrian collision sensor never fires in CARLA**
- **Dual verdict: ego ledger (RAW-HIT) + physics oracle**



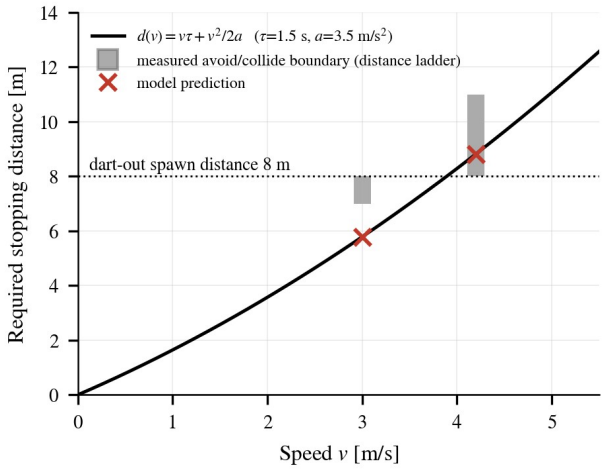
Results: Detection & Response

- **Detection 96–100% (19/20 conditions); 0 false alarms / 45 runs**
- **Exposure 238–275 m → 11–58 m; hard-fail decision ≤ 1.20 s**
- **Health bands separate by severity → tolerate / evacuate / defer**



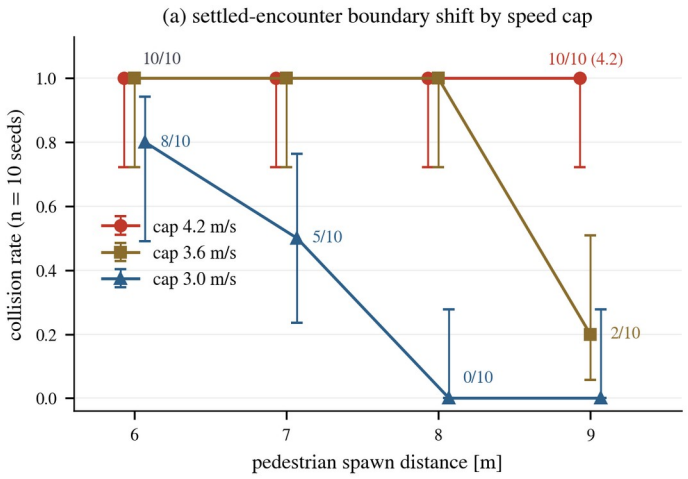
Results: Avoid/Collide Boundary

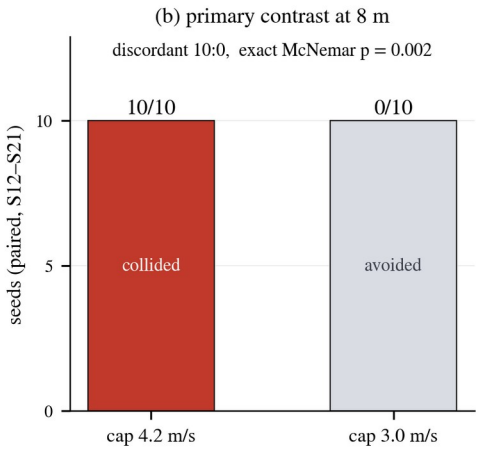
- **Boundary: 8–11 m at 4.2 m/s (model 8.8 m); 7–8 m at 3.0 m/s**
- **8 m dart-out: stock 10/10 collide vs settled cruise 8/8 avoid**
- **Transient encounters 2/2 collide → risk lives in the transient**



Results: Causal Cap Effect

- 120 pre-registered runs; settlement trigger — cap holds at encounter
- 8 m paired contrast: discordant 10:0, exact McNemar p = 0.002
- Boundary moves inward monotonically: beyond 9 m → about 7 m





Conclusion

- **Graded fault response — health score + MRM supervision, 1,862 cases**
- **Stop vs evacuate: a temporal trade-off; risk lives in the transient**
- **Settled low speed shifts the boundary (10:0, p = 0.002)**
- **Reusable verdict methodology: ledger + physics oracle**
- **Future: high-speed ODD, adaptive baselines, real vehicle**